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Spatiotemporal Integrated Gradients: Contribution Method for Video Data

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ABSTRACT

Unfinished.

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1. INTRODUCTION

Suppose you have a convolutional neural network (CNN) model trained to predict class labels (digits, people, objects) and that it surpasses humans on such classification tasks. If this model is designed in the fashion of popular networks like ResNet-18, then these models would have hierarchical representations of visual inputs. Thus, simple features such as classical receptive fields would appear in early layers and complex, specific features like faces or a person would appear in later layers. This hierarchical representation of visual stimuli in models is well-founded and closely represents human level organization of feature representation in the ventral visual stream.

The aforementioned logic assumes that the CNN has not learned a trivial look-up table for the inputs and their labels, which would make the following research questions and line of reasoning pointless. What representations within the CNN contribute to correct identification? What internal layer functions necessitate successful object categorization? And most importantly: what does each layer of the model see when given an input? Such line of questioning are at the core of explainable AI (XAI). XAI is closely related to the field of mechanistic interpretability, putting the 'science' in computer science and elucidating principles and laws of neural network computations. In simple terms, XAI and MechInterp have a shared goal of parsing how neural network models perform well-defined tasks.

1.1. *Background on Interpretability Methods*

For image classification tasks, interpretability tools like Gradient-weighted Class Activation Mapping (GradCAM) provides visual explanations for classification tasks. This method works by computing the gradients from an input with respect to the activations of the final convolutional layer, indicating the importance of each feature map. These gradients are averaged, summed, then thresholded to combine these features into a single map of important regions overlayed onto the original input. This method gives a visual representation of model choices per input image. The XAI community has expanded on this method of interpretability in vision networks with GradCAM++, ScoreCAM, and Integrated Gradients. The last of which I will expand upon for the rest of this document.

Integrated gradients (IGs), unlike CAM based methods, calculates a path integral from baseline to stimulus to any layer of interest. This is accomplished through making a path, collecting gradients, approximating the integral. This explains what part of the input contribute to model prediction. Further, using integrated gradients enables network contribution mapping: identifying channel specific attributions for convolutional layers. By leveraging the fundamental theorem of calculus: the integral and differentiation are inverse operations; integrated gradients asserts linearity, sensitivity, and completeness. Thus, integrating gradients through a neural network is a more rigorous attribution method for explaining model computation and functions.

1.2. *Motivation for Sequential Integrated Gradients*

We will expand the integrated gradient method in two ways: using (1) models trained to predict neuronal firing rates in response to (2) video inputs that the neurons observed. The following research questions can be asked: What does each layer of the neural network observe as determined by contribution maps from top contributing channels? What cell types are necessary for firing rate prediction?

The Integrated gradients—while powerful, works best on static images. This method accumulates gradients from baseline to the input, which allows the practitioner to glean what features of the input contribute most to model

output. The question lies in what a good baseline should be for calculating this integral. In one case integrating from black, or zero, would be analogous to seeing what features of the model are salient as the full image comes into view. It can be argued that this approach biases the explanation, thus, averaging the path across many baselines (e.g. variable starting points between black and gray) up to the full image gives a better estimate of the path attribution.

Applying this method to videos would require a per-frame IG computation or averaging across batches of frames—which would result in low computationally intensive and low quality estimations of the attributions. To sequentially compute integrated gradients accurately, we must compute a spatiotemporal integral. Only the first frame is integrated from zero to the full image and each subsequent frame uses the full image from the prior frame. This will allow us to find spatial contributions through time in the model.

[Continue with reference other time-aware uses of Integrated Gradients]

2. METHODS

2.1. *Standard Integrated Gradients*

The standard integrated gradients method computes attributions for static images by integrating gradients along a path from a baseline to the input. The attribution for feature i is given by:

$$\mathrm{IG}_i(x) = (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha \tag{1}$$

where F is the model, x is the input image, x' is the baseline (typically a black image), and α parameterizes the path from baseline to input

Algorithm 1 Standard Integrated Gradients (IG) Method

Require: F: model; x: input image; c: target neuron/class; x': baseline (e.g., black image); n: number of interpolation steps
Ensure: A: attribution map showing spatial importance

```
Initialize A ← 0 with same shape as x
Generate interpolation coefficients {α_1, α_2, ..., α_n} evenly spaced in [0, 1]
for each α in {α_1, ..., α_n} do
    Compute interpolated input: x_α ← x' + α × (x − x')
    Enable gradient tracking for x_α
    Forward pass: y ← F(x_α)
    Select target output: y_c ← y[c]
    Backward pass to obtain gradients: g_α ← ∂y_c / ∂x_α
    Store g_α
end for
Approximate path integral using trapezoidal rule:
    IG ← 0
for i = 1 to n − 1 do
    IG ← IG + 0.5 × (g_i + g_{i+1})
end for
Compute mean integrated gradient: IG_mean ← IG/n
Compute final attribution: A ← (x − x') ⊙ IG_mean
return A
```

2.2. *Sequential Integrated Gradients*

For video sequences, we introduce Sequential Integrated Gradients (SIG), which extends the standard IG method to handle temporal data. The key insight is to compute a piecewise path where each frame is revealed sequentially. For a video with T frames, the attribution for feature i in frame t is:

$$\mathrm{SIG}_{i(t)}(x) = \sum_{k=1}^{T} \mathrm{IG}_{i(t),k}(x) \tag{2}$$

$$\mathrm{SIG}_{i(t),t} = (x_{i(t)} - x_{i(t-1)}) \times \int_{(t-1)/T}^{t/T} \frac{\partial F(r^*(\alpha))}{\partial x_{i(t)}} \cdot T \, d\alpha \tag{3}$$

2.2.1. *Defining the Piecewise Path*

We define the path $r^*(\alpha)$ for $\alpha \in [0,1]$, where the total path is divided into T equal-length segments, each of length $1/T$.

Let $\mathbf{X}_t^{\text{partial}}$ be a "partial input" where the first t frames are set to their true values $(x_1, \ldots, x_t)$ and the remaining $T-t$ frames are set to their baseline values $(x'_{t+1}, \ldots, x'_T)$. We define $\mathbf{X}_0^{\text{partial}} = \mathbf{X}'$.

For the t-th segment, corresponding to $\alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right]$, we transition from the partially revealed input $\mathbf{X}_{t-1}^{\text{partial}}$ to the next partially revealed input $\mathbf{X}_t^{\text{partial}}$.

The scaled parameter β within the segment $[0,1]$ is defined as:

$$\beta = \alpha \cdot T - (t-1), \quad \text{for } \alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right] \tag{4}$$

The path $r^*(\alpha)$ is then defined by:

$$r^*(\alpha) = \mathbf{X}_{t-1}^{\text{partial}} + \beta \cdot (\mathbf{X}_t^{\text{partial}} - \mathbf{X}_{t-1}^{\text{partial}}) \tag{5}$$

Algorithm 2 Spatiotemporal Integrated Gradients (SIG) Method

Require: F: model; X: input video with T frames; c: target neuron/class; X': baseline video (all black frames); s: number of interpolation steps per segment

Ensure: Spatiotemporal attribution tensor $\text{SIG}_{i(t)}$ for each feature i in frame t

```
Initialize X' ← video of T black (zero) frames
Define partial input states: X_0^partial ← X', X_T^partial ← X
Initialize storage for frame-wise attributions SIG_{i(t)}
for t = 1 to T do
    Define segment boundaries on the integration path:
    α ∈ [(t−1)/T, t/T]
    Frames 1, ..., t−1 fixed at true values, frame t transitions from baseline to true value, frames t+1, ..., T remain baseline.
    Generate s interpolation steps {β_1, β_2, ..., β_s} where β_j = j/s
    for j = 1 to s do
        Construct partial video X^partial(j):
        For frames 1, ..., t−1: x_k^(j) = x_k
        For frame t: x_t^(j) = x'_t + β_j × (x_t − x'_t)
        For frames t+1, ..., T: x_k^(j) = x'_k
        Stack all frames as required by F (e.g., concatenate along channel dimension)
        Forward pass: y ← F(X^partial(j))
        Select target output: y_c ← y[c]
        Backward pass to obtain gradients: g_t^(j) ← ∂y_c / ∂x_t^(j)
        Store g_t^(j)
    end for
    Approximate integral for frame t using trapezoidal rule:
    IG_t ← 0
    for j = 1 to s − 1 do
        IG_t ← IG_t + 0.5 × (g_t^(j) + g_t^(j+1))
    end for
    Scale by segment length: IG_t ← T × IG_t / s
    Compute frame attribution:
    SIG_{i(t)} ← (x_{i(t)} − x'_{i(t)}) ⊙ IG_t
    Store SIG_{i(t)} in attribution tensor
end for
return Complete attribution tensor SIG containing spatiotemporal importance scores satisfying F(X) − F(X') = Σ_{t,i} SIG_{i(t)}
```

2.3. *Contribution Map Visualization*

After computing attributions using either IG or SIG, we can identify which channels in a given layer contribute most to the model's predictions. This enables visualization of what the model "sees" at different layers.

Algorithm 3 Contribution Map Generation

Require: Attribution tensor A (from IG or SIG), target layer L, number of top channels K, original input dimensions
Ensure: Visualization maps showing top contributing channels

```
attributions_L ← Extract attributions for layer L from A
if using SIG then
    ▷ Attributions have temporal dimension
else
    ▷ Attributions are static
end if
contributions ← []
for each channel k in layer L do
    score_k ← Σ_{i,j} |attributions_L[k, i, j]|    ▷ Sum over spatial dims
    Append score_k to contributions
end for
Sort contributions in descending order
top_channels ← Select top K channels
threshold ← mean(contributions) + std(contributions)
Filter channels: keep only those with score_k > threshold
for each selected channel k do
    activation_k ← Extract activation pattern from layer L during forward pass
    activation_resized_k ← Resize to input dimensions (bilinear interpolation)
    Normalize activation_resized_k to [0, 1] range
end for
Create tiled visualization grid of top channel activations
if temporal data then
    Create separate visualizations per time point or average across time window
end if
return Visualization maps
```

3. RESULTS

Unfinished

4. DISCUSSION

Unfinished

5. CONCLUSION

Unfinished

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B. AUTHORS' CONTRIBUTIONS

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C. CONFLICT OF INTEREST

The author declares no conflict of interest.

D. DATA AVAILABILITY

Empty

E. ETHICAL STATEMENT

Empty

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